METR — MODEL EVALUATION & THREAT RESEARCH · MARCH 2025

Measuring AI Ability to Complete *Long Software Tasks*

⚡ **Key Finding: 50% task-completion time horizon doubling every ~7 months**

AUTHORS
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SOURCE
arxiv.org/abs/2503.14499

BENCHMARK
170 tasks · 800+ human baselines · 2,529 hrs

METHODOLOGY

A New Metric: *50% Task-Completion Time Horizon*

1. **Assemble Diverse Task Suite**
   Curate 170 software & research tasks spanning 3 seconds to 8 hours of skilled human effort, drawn from HCAST, RE-Bench, and SWAA.

2. **Collect Human Baselines & AI Success Rates**
   Time skilled professionals on each task (800+ baselines, 2,529 hours total). Run AI agents on same tasks; measure binary success rates.

3. **Fit Logistic Model → 50% Horizon**
   Fit a logistic curve to AI success vs. human task duration. The **50% time horizon** = duration at which the model succeeds half the time. Intuitive and human-grounded.

Family	Length	Description
find_shell_script	3 seconds	Multiple choice: "Which file is a shell script?" Choices: "run.sh", "run.txt", "run.py", "run.md"
wikipedia_research	1 minute	Research simple factual information from Wikipedia and provide accurate answers to straightforward questions.
oxdna_simple	9 minutes	Detect and fix a bug in the input files for a molecular dynamics simulation using the oxDNA package.
munge_data	56 minutes	Write a Python script to transform JSON data from one format to another by inferring the conversion rules from provided example files.
cuda_backtesting	8 hours	Speed up a Python backtesting tool for trade executions by implementing custom CUDA kernels while preserving all functionality, aiming for a 30x performance improvement.

Fig. 2 — Methodology overview (Kwa et al., 2025)

TASK SUITE

170 Tasks Spanning *3 Seconds to 8 Hours* of Human Effort

HCAST
97
Diverse software tasks
1 min – 30 hrs

RE-Bench
7
Difficult ML research engineering, all 8 hrs

SWAA
66
Short atomic actions
1 sec – 30 sec

HUMAN BASELINES
800+ | 2,529 hrs

TASK NAME	HUMAN DURATION	DESCRIPTION
find_shell_script	3 sec	Multiple choice: "Which file is a shell script?"
wikipedia_research	1 min	Research simple factual information from Wikipedia
oxdna_simple	9 min	Detect & fix a bug in molecular dynamics simulation input files
munge_data	56 min	Write a Python script to transform JSON data by inferring conversion rules
cuda_backtesting	8 hrs	Implement custom CUDA kernels for 30× speedup of a backtesting tool

EXPONENTIAL GROWTH

GPT-2 (2 sec) → o3 (~1 hr 50 min) in *6 Years*

DOUBLING TIME

~207 days

≈ 7 months per doubling

MODEL FIT (R^2)

0.97

Exponential trend, 2019–2025

RECENT ACCELERATION

+20%

Faster growth rate 2023–2025 vs. full period

KEY MODEL MILESTONES

GPT-2 (2019)	~2 sec
GPT-3 davinci (2020)	~9 sec
GPT-3.5-turbo-instruct (2023)	~36 sec
GPT-4 (0314) (2023)	~5 min
Claude 3.5 Sonnet (2024)	~28 min
o1 (2024)	~39 min
Claude 3.7 Sonnet (2025)	~54 min
o3 (2025)	**~1 hr 50 min**

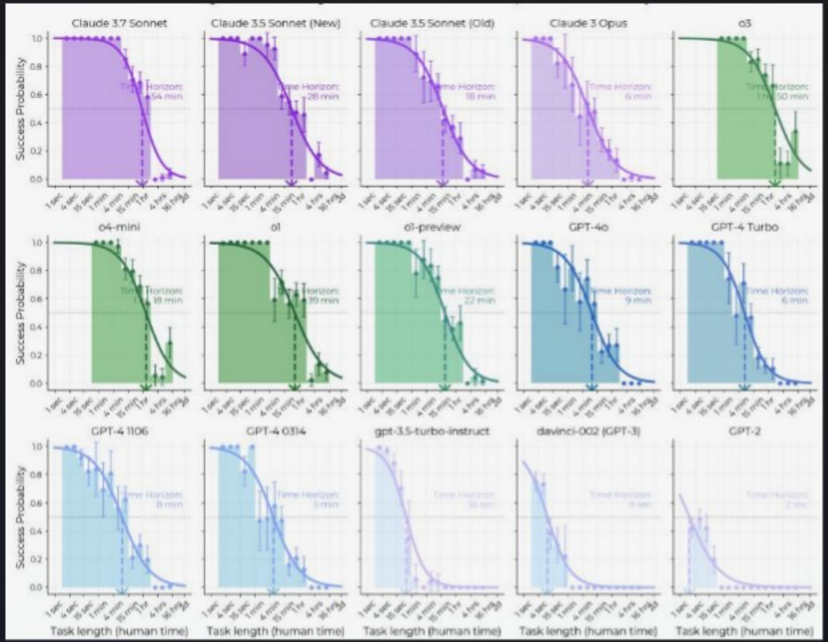

Fig. 1 — 50% time horizon vs. model release date · Doubling time ≈ 207 days · R^2 = 0.97

PER-MODEL PERFORMANCE

Success Probability Drops Sharply as *Task Length Increases*

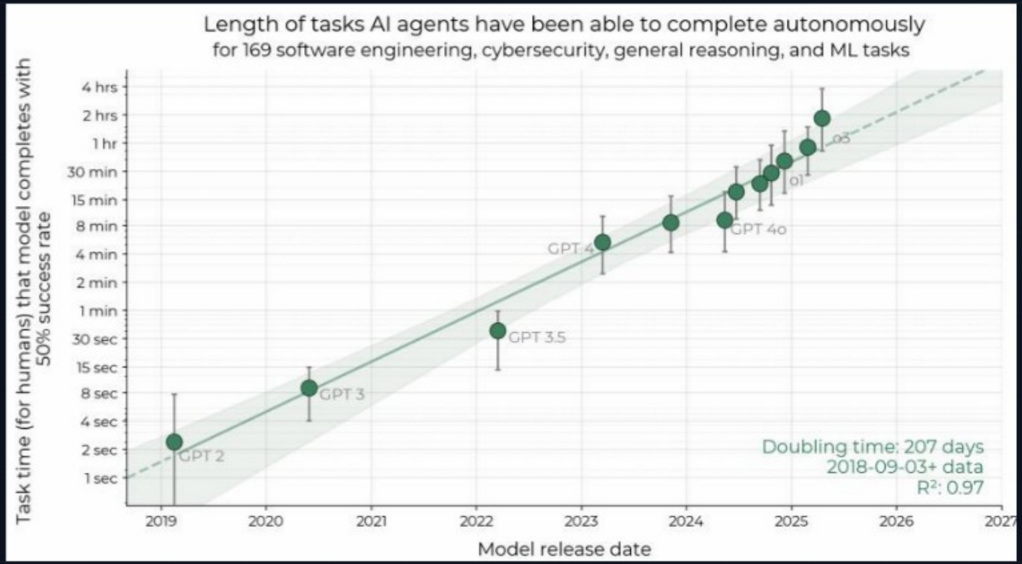

Fig. 3 — Success probability vs. task duration by model

Frontier Models Push Curve Right

o3 (~1 hr 50 min) and Claude 3.7 Sonnet (~54 min) extend the 50% horizon dramatically rightward compared to GPT-2's ~2 sec.

All Models Follow Logistic Decay

Success rate falls steeply as task duration grows. The logistic curve fits each model well, validating the 50% horizon as a stable summary statistic.

80% Horizon ~5× Shorter

The 80% time horizon (tasks completed 80% of the time) follows the same trend but horizons are roughly 5× shorter than the 50% estimates.

Messy Tasks Underperform

Performance is notably lower on unstructured, less well-defined tasks — a key external validity caveat for real-world deployment.

DRIVERS & LIMITATIONS

Reasoning, Tool Use, and *Error Recovery* Drive Growth

CAPABILITY DRIVERS

🧠 **Improved Logical Reasoning**
Models better decompose multi-step problems and maintain coherence across long task horizons.

🔧 **Better Tool Use**
More reliable invocation of shell, code execution, file I/O, and search tools without human scaffolding.

🎯 **Reliability & Self-Awareness**
Frontier models more accurately gauge their own uncertainty and avoid catastrophic irreversible mistakes.

↩️ **Ability to Adapt to Mistakes**
Models increasingly recover from errors mid-task rather than compounding failures or giving up.

LIMITATIONS & CAVEATS

⚠️ **Poor Performance on Messy Tasks**
Unstructured, ambiguous, or "messy" real-world tasks remain a significant weak point not fully captured in the benchmark.

🔍 **External Validity Uncertain**
Benchmark tasks may not perfectly represent average segments of real-world intellectual labor across job distributions.

📊 **Supplementary Evidence Mixed**
Additional experiments found little evidence that trends are slower on more realistic tasks — but the question remains open.

FORWARD EXTRAPOLATION

Month-Long Task Automation *Projected 2028–2031*

CAPABILITY TRAJECTORY

- 2019 — **GPT-2 — ~2 seconds**
  Baseline; near-zero autonomous task capability
- 2023 — **GPT-4 — ~5 minutes**
  Growth rate accelerates ~20% above the 2019–2025 mean
- 2025 — **o3 — ~1 hr 50 min**
  Current frontier: autonomous completion of ~2-hour software tasks
- 2028–2031 — **Projected: ~1 Month (167 work hrs)**
  Naive extrapolation of current doubling rate; implies automation of many month-long software projects

EXTRAPOLATION CAVEATS

Naive extrapolation only

The 2028–2031 forecast assumes no change in doubling time. The actual pace may accelerate or plateau depending on algorithmic and hardware developments.

~20% acceleration since 2024

The 2023–2025 growth rate is approximately 20% faster than the full 2019–2025 rate, suggesting the lower bound of 2028 may be more likely.

External validity unknown

Benchmark tasks skew toward well-defined software tasks. Real-world month-long projects involve coordination, ambiguity, and stakeholder management not yet tested.

167 work hours = ~1 month

The threshold is defined as 167 work hours (≈ 4.2 full work weeks), representing the scale of many software projects that currently require dedicated human teams.

KEY TAKEAWAYS

Exponential Autonomous Capability Growth Demands *Proactive Safety Governance*

01 Robust, Intuitive Metric
The 50% time horizon provides a stable, human-grounded capability measure with R^2=0.97 over 6 years — suitable for policy tracking and safety thresholds.

02 ~7-Month Doubling With Possible Acceleration
The ~207-day doubling time since 2019 appears to be accelerating since 2024, with the recent rate ~20% faster than the historical average.

03 Frontier Capability: ~2 Hours Today
o3 (2025) can autonomously complete tasks taking skilled humans nearly 2 hours — a milestone with direct implications for software automation and oversight requirements.

04 External Validity Remains Open
Tasks may not perfectly represent real-world intellectual labor distributions. Limitations include poor performance on unstructured tasks and uncertain generalization.

05 Month-Long Automation by 2028–2031 — Governance Must Lead the Curve
Naive extrapolation predicts AI reaching a 1-month time horizon (167 work hours) between mid-2028 and mid-2031. Safety frameworks, oversight mechanisms, and policy structures need to be developed well before this capability threshold is reached.